Abstract

Handwritten Digit Recognition using Neural Networks by Sangini Agrawal, Naviotech Solution AI Internship (September 2026 batch)

Handwritten digits appear on cheques, postal envelopes, examination sheets and paper forms, and reading them automatically is a classic problem in pattern recognition. This project builds and evaluates neural networks that recognise handwritten digits (0–9) from 28 × 28 grayscale images of the MNIST dataset of 70,000 labelled samples. The pixel values were scaled to the range 0–1 and 6,000 of the 60,000 training images were held out as a stratified validation set, so that the 10,000 test images were used only once, for the final evaluation.

Two models were implemented in TensorFlow/Keras. The main model, a dense neural network with two hidden layers of 128 and 64 ReLU neurons, 20% dropout and a softmax output (109,386 parameters), reached a test accuracy of 97.93%. An improved convolutional neural network with two convolution–pooling blocks (34,826 parameters) reached 99.19% with a macro F1-score of 0.9918, making 81 errors on 10,000 images. Both models exceed the project target of 95% and outperform logistic regression (92.45%), k-nearest neighbours (96.99%) and a random forest (96.96%). Most of the remaining errors are on digits with similar shapes, such as 4 and 9 or 7 and 2, and most are made with low confidence.

A prediction system lets users test their own handwriting by drawing on a canvas in Google Colab, uploading a photograph, using a command-line tool or a browser demo. User images are prepared the same way as MNIST digits (inverted, cropped, fitted into a 20 × 20 box and centred). With this preprocessing the CNN read all eleven sample images correctly. On 30 harder photo-style images it got 16 right, compared with only 4 with a plain resize, which shows both why preprocessing matters and that small digits on noisy paper are still difficult.

Keywords: handwritten digit recognition, MNIST, neural networks, convolutional neural networks, deep learning, TensorFlow, Keras, image classification